

Few-Shot Learning with Pretrained Visual Embeddings of Eye-Tracking Patterns for Autism Detection

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Background: Autism Spectrum Disorder

- Autism Spectrum Disorder (ASD) is a pervasive developmental disorder characterised by a set of impairments including social communication problems.¹
- ASD has been considered to affect about 1% of the world's population (US Dep. of Health, 2018).²
- The hallmark of autism is an impairment of the ability to make and maintain eye contact.³

¹ L. Wing, and J. Gould, "Severe Impairments of Social Interaction and Associated Abnormalities in Children: Epidemiology and Classification". Journal of Autism and Developmental Disorders, 9(1), pp.11-29, 1979.

² U.S. Department of Health & Human Services. Data and statistics | autism spectrum disorder (asd) | ncbddd | cdc, 2018. URL: <https://www.cdc.gov/ncbddd/autism/data.html>.

³ Coonrod, E. E. and Stone, W. L. (2004). Early concerns of parents of children with autistic and nonautistic disorders. Infants & Young Children, 17(3), 258–268.



Background: Eye-Tracking Technology

Screen-based eye trackers



Glasses

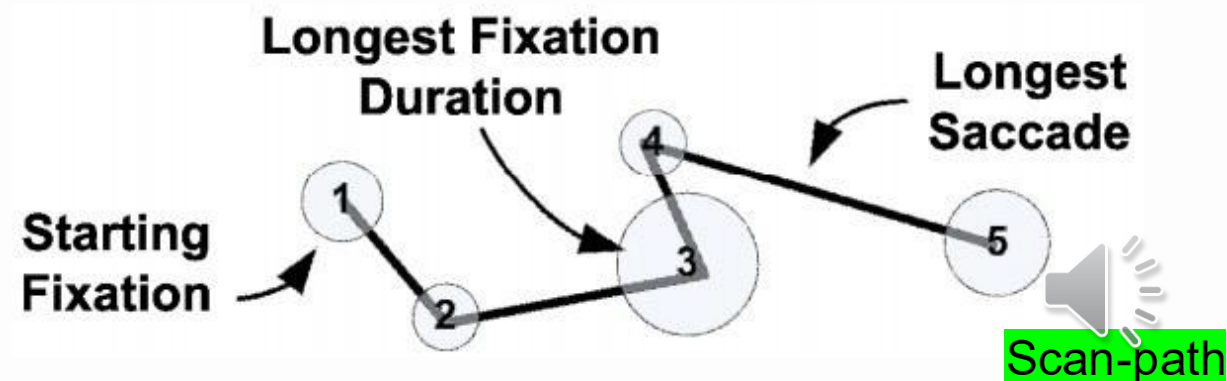


Image Source: <https://imotions.com/blog/eye-tracking/>

J.H. Goldberg, and J.I. Helfman, "Visual scanpath representation", In Proceedings of the 2010 Symposium on Eye-Tracking Research & Applications, ACM, 2010, pp. 203-210.

Motivation

Our Goal:

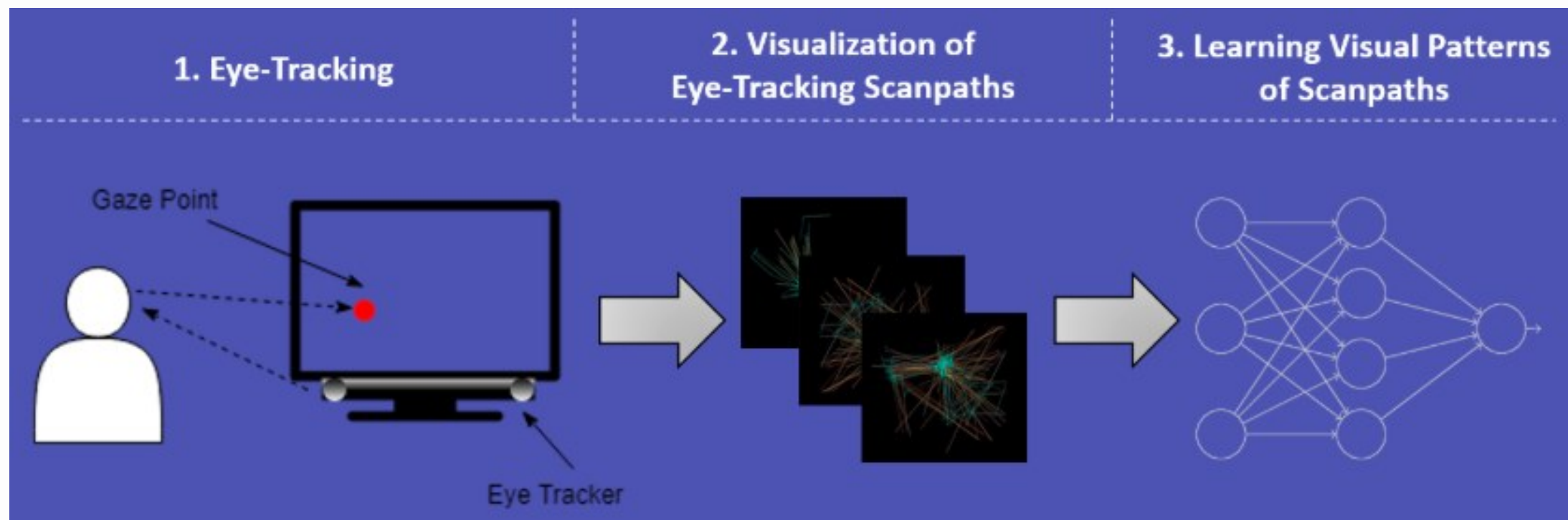
- Detecting ASD-diagnosed individual in eye-tracking data.
- Addressing the challenge of limited or scarce eye-tracking data.

Key Idea:

- Compactly render eye movements into an image-based format while maintaining the dynamic characteristics of eye motion.
- As such, the classification task could be approached as image classification.
- Apply few-shot learning to address the challenge of limited data.



Our Earlier Work (HEALTHINF 2019, 2022)



Carette, R., Elbattah, M., Cilia, F., Dequen, G., Guérin, J., & Bosche, J. (2019). Learning to predict autism spectrum disorder based on the visual patterns of eye-tracking scanpaths. *In Proceedings of the 12th International Conference on Health Informatics (HEALTHINF)*.

Elbattah, M., Guérin, J., Carette, R., Cilia, F. & Dequen, G. (2022). Vision-based approach for autism diagnosis using transfer learning and eye-tracking. *In Proceedings of the 15th International Conference on Health Informatics (HEALTHINF)*.

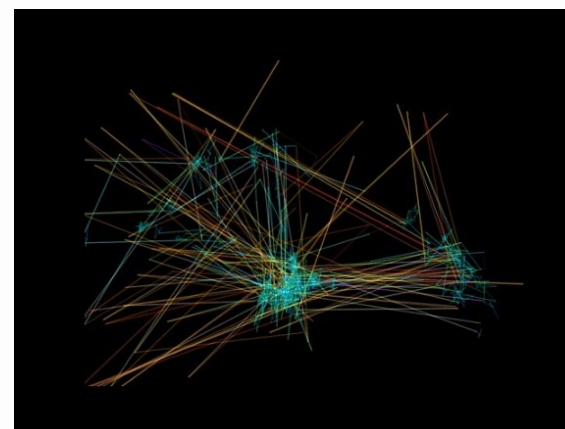
Data Description

Number of Participants (ASD, TD)	59 (29, 30)
Gender Distribution (M, F)	38 ($\approx$ 64%), 21 ($\approx$ 36%)
Age (Mean, Median) years	7.88, 8.1
CARS Score (Mean, Median)	32.97, 34.50

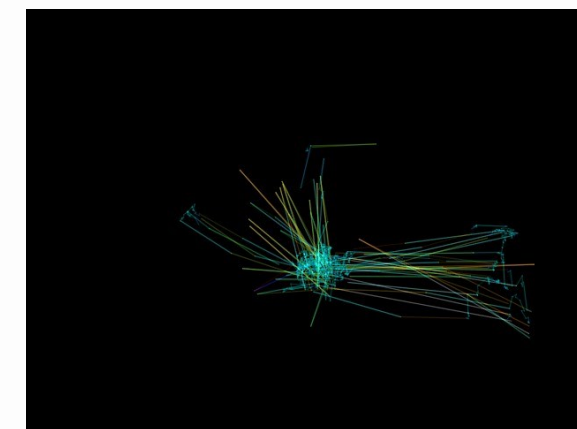


Image Dataset Description

- 547 images: 328 (Non-ASD), 219 (ASD)
- Image dimensions: 640x480



ASD

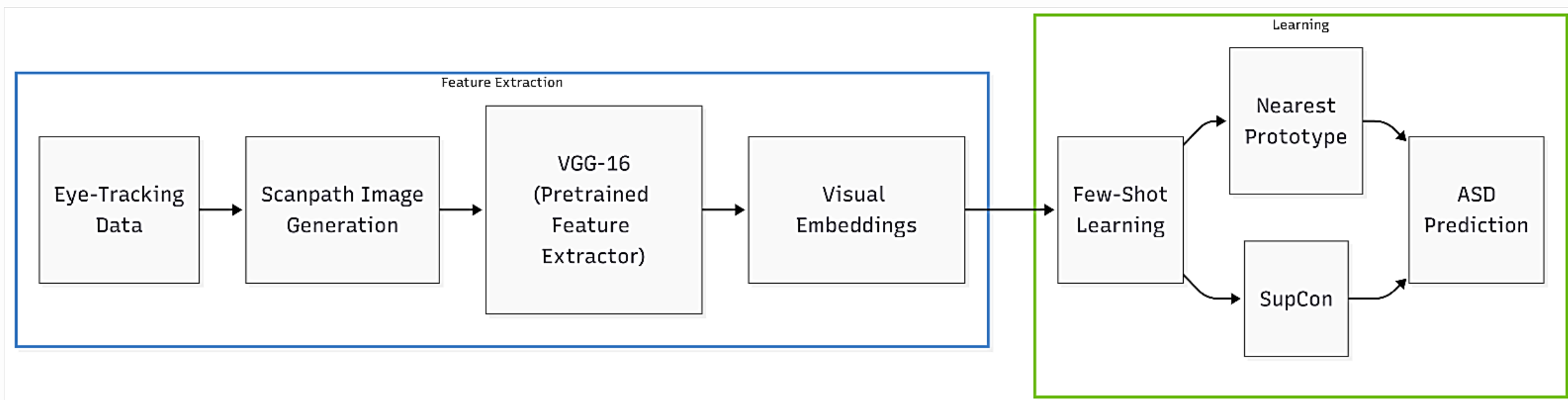


Non-ASD



Carette, R., Elbattah, M., Dequen, G., Guérin, J. L., & Cilia, F. (2018, September). Visualization of eye-tracking patterns in autism spectrum disorder: method and dataset. In *2018 Thirteenth International Conference on Digital Information Management (ICDIM)* (pp. 248-253). IEEE.

Overview of Methodology



Results — Nearest Prototype

Shots	AUROC (± SD)	AUPRC (± SD)	Precision (± SD)	Recall (± SD)
1-shot	0.781 (±0.092)	0.695 (±0.079)	0.377 (±0.031)	0.874 (±0.086)
3-shot	0.777 (±0.082)	0.687 (±0.075)	0.457 (±0.029)	0.739 (±0.155)
5-shot	0.782 (±0.088)	0.697 (±0.080)	0.468 (±0.026)	0.754 (±0.133)
10-shot	0.779 (±0.094)	0.689 (±0.083)	0.477 (±0.027)	0.739 (±0.141)



Results — SupCon

Shots	AUROC (± SD)	AUPRC (± SD)	Precision	Recall
1-shot	0.764 (±0.063)	0.660 (±0.068)	0.662 (±0.063)	0.507 (±0.164)
3-shot	0.765 (±0.059)	0.669 (±0.056)	0.629 (±0.049)	0.507 (±0.139)
5-shot	0.777 (±0.050)	0.675 (±0.050)	0.630 (±0.028)	0.525 (±0.136)
10-shot	0.750 (±0.072)	0.650 (±0.077)	0.641 (±0.018)	0.480 (±0.154)



Comparative Summary

- **Nearest Prototype**
 - High recall
 - Effective when minimising missed ASD cases
- **SupCon**
 - Higher precision
 - Reduces false positives



Conclusions and Limitations

- Both few-shot methods operate effectively with as few as 1–5 shots.
- Complementary strengths: sensitivity vs. precision.
- Small dataset.
- Scanpath images lose some temporal dynamics.
- It is conceived that the scarcity or imbalance of datasets could make such FSL approaches attractive for further investigation.



Thank You!

Please feel free to contact me for any opportunity for collaboration:

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